

Data-Driven Platforms for Aging in Place

1. General information

1.a Applicants

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant/Co-applicant]
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 Secondary Convergence theme (if applicable): Technology Supported Transitions in Health(care).
 Keywords: *Aging in Place, data-driven platforms, lived experiences, organizational readiness, sociotechnical systems, networked innovation*

1.b Summary

A rapidly aging population is putting unsustainable pressure on Europe's healthcare systems. In response, governments are encouraging adults to age at home. This policy to encourage "aging in place" (AiP) aligns with most aging adults' preferences. AiP sustains familiar activities, benefitting overall quality of life.

Simultaneously, AiP is characterized by functional and social challenges. With fewer hands-on resources, healthcare is undergoing a technology-transition to address these challenges. Technologies abound, offering mobility and communication support, sensory aids, health and activity monitoring. This overload of technological solutions is overwhelming for aging adults and caregivers. We argue this can be managed effectively through data-driven platforms. Data-driven platforms are digital networks that can personalize care by connecting people, products, and services through data.

Data-driven platforms signify a promising transition in supporting aging in place, which requires technology capable of adequately anticipating aging people's unique and changing needs. However, this transition could profoundly transform the experience of care, which is typically an intimate embodied practice providing meaningful human interaction. Furthermore, this transition will shift roles and responsibilities in care networks and require new organizational structures and financing schemes.

Innovative interdisciplinary research is required to create effective, sustainable and responsible data-driven AiP platforms. We will investigate opportunities and bottlenecks in data-driven AiP platforms from the perspectives of (1) aging adults, (2) care networks, and (3) data-driven platform technologies. This provides our foundation for developing

frameworks and strategies for the data-driven AiP platforms of the future. It is our vision that these platforms should function as a friend that grows along with the aging adult, orchestrating an ecosystem of people, products and services to help them age in place. Our Flagship will launch an Aging in Place Knowledge and Innovation Center that unites the disciplines necessary to realize our vision.

1.c Public Summary (max 300 words)

Imagine a future in which senior citizens are able to live happy, active and socially meaningful lives in their own environment, despite functional decline and chronic medical conditions. What if digital technology could make this future actual, supporting the needs and challenges of people who age at home in empowering ways. We believe data-driven platforms have the potential to do just that.

Data-driven platforms are digital networks that can personalize care by connecting people, products, and services through data. Data-driven platforms are promising for providing aging people with targeted resources, building on technology that can adequately anticipate aging people's unique and changing needs. However, they also raise important concerns. Our multidisciplinary Flagship will tackle these issues heads-on. For example, we will investigate the experience of care by aging adults to learn how data-driven platforms can preserve care's human dimension. As data-driven platforms will also transform practices, roles and responsibilities in existing care networks, we will also investigate how care providers manage this transition. Data-driven platforms also collect and process highly personal and sensitive data, so we will develop effective platform mechanisms that treat these data responsibly.

It is our vision that data-driven platforms supporting aging adults should function as a friend that grows along with the aging adult, orchestrating a connected system of people, products and services to help them "age in place." To realize this vision, our Flagship will launch an Aging in Place Knowledge and Innovation Center to pave the road for a future in which data-driven platforms can effectively, sustainably and responsibly support adults aging in place.

2. Relevance to Convergence Health & Technology mission

2.a Contribution to H&T mission and fit with H&T theme(s)

Aging in Place³ (AiP) is "the ability to live in one's own home and community safely, independently, and comfortably, regardless of age, income, or ability level." AiP is a complex journey with many challenges for aging adults and their caregivers^{1,17, 32}. A growing number of technologies are available to support AiP¹⁵. This abundance in technological solutions is often overwhelming for aging adults and their care networks^{8,19}. The result is a suboptimal use of the available technologies paired with a pressing concern how to orchestrate the delivery of the right technologies to the right person at the right time.

Data-driven platforms, which are digital networks connecting people, products, and services through data, can unite and personalize available yet disconnected local technologies³³. As such, these platforms have the power to provide individually tailored

and proactive support by cleverly orchestrating available care resources. Our Flagship's aim is to identify what makes data-driven platforms successful in supporting AiP¹ and the overall Quality of Life (QoL) of aging adults while simultaneously relieving an overburdened healthcare system⁴. Our research therefore contributes to several of the Convergence **Health & Technology mission and themes**:

We contribute to the **lifelong health and QoL** of aging adults by contributing to effective, sustainable and responsible support for AiP. There is broad scientific support that AiP improves the QoL of aging adults by sustaining familiar activities, routines and connections^{14,27}. That said, AiP also comes with challenges showing up in unique ways over time. Following the ethical principle 'nothing about us without us,' our research adopts an emancipatory approach prioritizing the needs of aging adults who age in place. On this basis we identify how data-driven platforms should be designed and embedded, such that they optimally match people's needs with the wide range of resources capable of supporting them. This will contribute to the Convergence commitment to **individually tailored and preventive care**²⁴. Crucially, data-driven AiP platforms should be accessible for all aging adults²⁵. We therefore adopt an **inclusive** approach, including people in the research with different cultural and socioeconomic backgrounds and varying levels of digital skills²¹. Two of our partners, Genero and NOOM, help safeguard this inclusivity.

Our Flagship also aligns with the theme of **improving health journeys**, since we understand AiP as a journey in which different needs for support increase over time^{7,11,32}. After the age of 65, people typically experience functional decline; difficulties with hearing, seeing, and mobility, while also often coping with one or more chronic diseases, such as diabetes, cardiovascular diseases or COPD. This puts burdens on daily life as aging adults increasingly need medical management and adherence to a variety of medications. Aging adults will also frequently experience significant changes in their social networks, with children establishing their own families or friends and relatives passing away²⁹. Aging adults, their family members and friends, or their formal caregivers may notice help is needed. By providing access to the right care, supported by the right technology, at the right time, data-driven platforms can support AiP precisely as an individual journey. This also postpones (costly) admissions to healthcare facilities, such as a nursing home. As such, data-driven platforms are likely to become instrumental for healthcare systems.

Our Flagship also connects with the theme, **technology supported transitions in healthcare**. With formal care resources becoming scarce, it is essential to develop technological interventions capable of meeting the variety of challenges and needs of aging adults^{12,15}. Data-driven AiP platforms can meet this requirement by design, since they are able to use the data generated by their users to effectively personalize and anticipate the resources and services their users need. Crucially, though, a sustainable and responsible transition towards data-driven AiP platforms should also anticipate their impact on how aging adults experience care, since care typically consists of hands-on embodied interactions, which often play an important social role in aging people's lives. Furthermore, this transition will shift roles and responsibilities in care networks and require new organizational and financing schemes. Finally, this transition will affect data-sharing practices between aging adults and their care networks². Our Flagship will investigate how these transitions can be made responsibly.

2.b Synergy between disciplines

One key aim of our Flagship is to identify bottlenecks in existing data-driven AiP platforms that if left untouched, will undermine the development of effective, sustainable and responsible platforms in the future. To identify these bottlenecks, we will bring together medical, organizational, technical, social, ethical, and economic methods, with which we will investigate data-driven AiP platforms from three perspectives:

Perspective 1: Aging Adults

Public Health pursues the prevention of diseases and promotion of health in society. It studies, among others, how health technology can effectively support aging adults.

Geriatrics offers a medical perspective on the aging process, as it aims to promote health by preventing and treating diseases and disabilities in aging adults. **Anthropology** studies human behavior and experience as embedded within societies and cultures. We use it to articulate the life-worlds of aging adults and the meaning they attribute to AiP, including the experience of traditional versus digitally mediated care¹³. Measuring the experience of AiP is a necessary step to develop effective data-driven AiP platforms. We will use our expertise in **Data Science**, an interdisciplinary field that models and quantifies phenomena, to generate data-driven insights that capture the various needs and experiences of aging adults based on connections across a variety of data sources⁵.

Perspective 2: Care Networks

Data-driven AiP platforms will, to varying degrees, transform the functioning and organization of (in)formal care networks²⁸. We use **Health Policy and Management** to understand the impact of data-driven AiP platforms on the governance, financing and organization of healthcare systems. We will also use **organizational Studies** to assess how Data-driven AiP platforms can relieve the burden of healthcare organizations while also demanding new ways of providing care. This may require new skills and shifts in responsibilities²³. Lastly, **General Practitioners** (GPs) represent a distinct profession of medical doctors that play a key role in coordinating care around an aging person. Data-driven AiP platforms provide GPs with efficient ways to monitor the health and wellbeing of their patients, with tools for the management of chronic conditions, and with information to timely and accurately intervene in the aging adult's health journey.

Perspective 3: Data-driven Platforms

We also aim to understand the optimal design and use of data-driven platforms^{5,9}. **Design** is an interdisciplinary field focusing on creating solutions for societal problems by integrating knowledge about people, technology and organizations. Design offers methods and techniques to involve aging adults and other stakeholders in the co-creation of data-driven AiP platforms, increasing acceptance of these platforms in daily life. **Human-Computer Interaction** researches the design and use of computer technology. Data-driven platforms can be used through different interfaces (screen-based, speech-based, etc.) that need to be made natural and intuitive. This holds especially for health care systems that need to be fully inclusive. **Information systems**, which focuses on sociotechnical systems designed to collect, process, store, and distribute information, is essential for analyzing and designing data-driven AiP platforms that reconfigure care networks, products and services as supportive structures. Additionally, **Artificial Intelligence** will be used to analyze data collected by platforms and to make predictions about the kind of support and the technologies needed.

By developing a robust multidisciplinary understanding of data-driven AiP platforms, we learn what makes them effective and sustainable: effective insofar as they perform as intended by its design and sustainable insofar as they become genuinely embedded in the lives of aging adults and their wider care networks. Equally important is that data-driven AiP platforms are developed empathetically and responsibly. To this end, our Flagship utilizes its expertise in **Ethics of Technology**, which addresses the normative ethical issues arising in the use and development of technology. Throughout our Flagship, we will rely on this field to offer a normative perspective on the impact that data-driven AiP platforms will have on the experience and delivery of care and on the acquisition and use of data obtained through data-driven platforms.

2.c Long term sustainability plan

We will establish the “**Aging in Place Knowledge and Innovation Center**” whose aim is to investigate the potential of future data-driven platforms for supporting AiP. Our center is **sustainable in the long-term**, given the urgent societal problem of a rapidly aging population putting increasing pressure on healthcare systems across the globe. The use of technologies to support AiP, personalized through data-driven AiP platforms, will remain essential for many years to come.

The center will position itself at the forefront of the scientific and practical developments in the field, e.g. the development of an Internet of Things (IoT), advancements in natural interaction technologies, artificial intelligence and machine learning and their deployment in homes and in organizations¹⁶. The center will stimulate the exchange of knowledge among researchers, aging adult communities, formal caregivers, technology developers, government, and funding agencies through regular meetups and workshops (see section 4d for details). This will enable all stakeholders to contribute to more effective, sustainable and responsible AiP platforms.

To ensure **long-term sustainability** of the center, we will create strong visibility and reputation in the Netherlands and internationally during the first five years (see section 2d). This positions us to join upcoming (large size) funding schemes, including NWO and NWA on the crossover of the Healthy Aging and AI themes. We also expect new Ambient Assisted Living programs to be started up by the European Commission. This funding program aims to create better QoL for aging people through ICT-based products, services and systems and strengthen industrial opportunities of healthy aging innovation. The center will also obtain revenue through consultancy, training and engagement in innovation workshops and projects (see section 2d).

We identify **three potential risks** for our project. The **multidisciplinary** nature of the Flagship carries the risk of becoming fragmented. To mitigate this risk we will create a physical presence, next to a clear online presence, where researchers involved in the center meet regularly. This will also be the central meeting place for PhD's, postdocs and supervisors. Studying AiP entails the collection of **sensitive data**. By cooperating with organizations specialized in the storage and exchange of such sensitive data, we ensure compliance with relevant regulations. In fact, we aim to exceed those standards from a security and ethical perspective. Data stewards at our universities will be consulted in each step of the way. The third risk stems from the fact that AiP cannot be studied in the lab and **collaborations with providers of care and (platform) technologies** are needed. In these collaborations the intellectual property rights of all the parties involved as well as the

academic freedom to publish research findings need to be safeguarded. However, the TUD has ample experience with such collaborations, we will work with their legal department to craft the necessary agreements.

2.d Visibility, attracting talents and new partners

We will promote national and international visibility for our Flagship in various ways. We will create a professional online presence of the center for easy access. For the scientific community, we will publish and present our work within academic communities connected to the multidisciplinary nature of technological transitions in AiP. Examples of relevant publication outlets and conferences, corresponding to our three research-perspectives, are: Aging adults (Gerontology Journal; Age and Aging Journal, International Conference on Integrated Care); Care networks (Organizational Behavior in Health Care, European Health Management Association, European Group of Organizational Studies); Platform perspective (Computer-Human Interaction, IEEE Pervasive Computing and Communications, Interacting with Computers Journal). Our normative-ethical findings on the responsible development of data-driven AiP Platforms will be submitted to journals such as Science and Engineering Ethics; Ethics and Information Technology, and the Journal of Responsible Innovation.

The research collaboration with our partners will also result in concrete deliverables, such as prototypes and design demonstrators that we showcase to healthcare and design communities, health-tech industry, and policy makers. Examples of such venues are the international conferences on Design4Health and Designing Interactive Systems (DIS).

We further create visibility through regular scientific meetings organized by the center and in collaboration with broader Convergence and Medical Delta initiatives. Here we will invite our key stakeholders, including health insurance companies, care organizations and the Dutch Ministry of Health, Welfare and Sport (VWS), and allow new partners to join. We create a broader societal visibility with the help of our partner Omroep MAX; the established Dutch public broadcast corporation that develops television and online media for a large Dutch audience aged 50+. By reserving regular airtime for our center in talk shows, updates will be given about the achievements of our unfolding research agenda. This creates awareness about our research and the importance of AiP with the general public.

We have dedicated targeted work packages to developing and attracting talent (WP6) and aligning the center's research activities with education (WP5), thus anchoring these goals firmly into the center's core activities (see section 3b).

The center will also be open to and of interest for new partners for multiple reasons. Concerning knowledge transfer, the center provides consultancy about the design, implementation and management of data-driven AiP platforms. It will also benchmark the quality of existing data-driven AiP platforms and give advice on ways to improve them. The center will also create value by engaging new, but also existing partners, in workshops and projects, with partners working together with researchers and students on platform-related innovations. Furthermore, we intend to offer practical-oriented training to aging adults, informal caregivers and healthcare professionals on how to engage with data-driven platforms on a variety of technical, organizational and usage topics. Next to the funding

strategy explained in section 2c, these activities can generate additional revenue for the center from the healthcare sector and (high-tech) industry.

2.e Long term societal impact

Technology capable of supporting AiP will be increasingly important in the future, due to an overburdened healthcare system. As such, our center is expected to make a **significant contribution in supporting and promoting a healthy aging society**. Data-driven platforms offer an effective deployment of technology by listening to and anticipating the needs of aging adults, situated within their specific social and physical contexts. Such platforms can configure care networks and technologies as supportive structures around aging adults over time. Through hands-on collaborations with our knowledge dissemination partners, the center will further raise awareness among Dutch aging citizens about the available technologies for AiP.

The innovation processes that our research informs and enacts, align academic knowledge development with concrete innovations, allowing for a **dual valorisation strategy**. Firstly, the **knowledge** that is developed about data-driven AiP platforms can be used in industry, healthcare and social care to make platforms more effective, sustainable and responsible. Our center will provide access to this knowledge through consultancy, training and publications. Secondly, our innovation workshops and projects lead to actual **innovation** both in the form of incremental improvements to existing data-driven platforms (TRL7-9) as well as radically new design proposals for data-driven AiP platforms that can traverse within the center from TRL3 (proof of concept) towards TRL6 (design demonstrators). If the opportunity for a start-up presents itself, this can be supported through incubator programs such as YESDelft! and the Erasmus Center for Entrepreneurship.

3. Flagship proposal

3.a Ambitions

The major aim of our Flagship is to identify what makes future data-driven platforms successful in proactively providing support tailored to each adult aging in place. We will investigate bottlenecks and opportunities of existing data-driven platforms for aging in place to make them more effective, sustainable and responsible. To make our Flagship visible and accessible, we will launch the “Aging in Place Knowledge and Innovation Center”. The Flagship builds on rapid developments in technology to relieve overburdened care systems impacted by increasingly aging populations. The center will increase the innovation strength of healthcare by developing and sharing strategies on how to best design, implement and manage data-driven AiP platforms. A detailed overview of how our ambitions **align with the Convergence Health & Technology** research agenda is provided in section 2a.

The **innovative** aspect of our Flagship is the focus on data-driven platforms that promote the QoL of aging adults as a proactive and preventive approach to health. Rather than studying singular solutions for specific problems, we focus on platforms as a means of providing access to multiple products and services^{4,6}. Their data-driven nature enables them to optimally match care provision to the specific needs aging adults experience while aging in place. This broad focus makes the research **challenging** and prompts us to (1) create an understanding of human experience and behavior in ways that make it actionable in data-driven platforms yet preserves the richness and nuances that characterizes daily

living, (2) strategize for the complexity of matching the heterogeneity of aging people's needs with the breadth of opportunities to orchestrate different products and services, (3) safeguard the responsible and ethical access to data by formal and informal caregivers, (4) support a smooth transition of care networks in which actors may take on new responsibilities, once data-driven AiP platforms are in place.

Our Flagship will adopt an **innovative research approach** to investigate data-driven AiP platforms as participatory sociotechnical systems. This requires multidisciplinary research to analyze data-driven AiP platforms as they pertain to the needs and experiences of aging adults, as part of care networks, and as sociotechnical manifestations (see section 2b).

Challenging for this project is to synthesize these different perspectives in the creation of an integrated, emancipatory understanding of these platforms (for designing, implementing and managing them). Another **challenge** is working with the consortium partners to scope the areas of improvement that benefit from the participatory research and innovation process. With the high levels of expertise and experience that possess, we will be able to tackle these challenges successfully.

3.b Aims and feasibility

The feasibility of our Flagship project is ensured through the creation of a **multidisciplinary research team**: 6 PhDs and 3 Postdocs with supervisory teams consisting of researchers with different backgrounds and training (see workplan 3c). The three participating universities and their research groups provide the necessary expertise and skills to investigate data-driven AiP platforms from different disciplines, (section 2b). The multiplicity of research projects being conducted by researchers, partners and students in concert, requires careful coordination and alignment among PhD and Postdoc projects through WPs to **mitigate the risk** of fragmentation and foster synergies between projects. This is the task of WP8 (section 3c).

The **budget** (Appendix I) covers costs for the research team to carry out research activities. Budget is also reserved for research with our partners. This includes a budget to ensure an inclusive representation of aging adults, so that those with a lower SES will also be able to participate. Our budget covers research activities conducted within our universities, with local community centers, in people's homes or embedded in organizations of caregivers. Budget is also reserved for building prototypes and demonstrators. This includes budget for physical materials and electronics to build interactive products or adapt existing ones; purchasing software applications to build user-interfaces; programming tools for developing machine learning algorithms and APIs; and budget for installing and maintaining ICT infrastructures for cloud services facilitating data exchange across local and global networks. A financial buffer is created to **mitigate the risk** of unforeseen costs across these activities.

For the duration of the Flagship's first five years, multiple **grant proposal opportunities** will be pursued. To support the Flagship with additional funding, we are already submitting a Dutch NWO proposal on Access to Care in the Living Environment (KIC). We are in dialogue with the Dutch Ministry of VWS for funding through their innovation funds. In addition, we are looking into joining the "Academische Werkplaatsen" an initiative of VWS in collaboration with ZonMW. We also expect revenues to come in through the center's valorization activities (consultancy, training, workshops, projects).

3.c Work plan

Our **long-term work plan** is structured according to a **research and innovation process** that involves the team and partners collaborating in field studies, co-creation, experiments and targeted interventions. These collaborations will produce novel knowledge on data-driven AiP platforms and platform innovations. The Flagship consists of 8 Work Packages (WPs). WPs 1-4 are research-oriented, with the first three WPs corresponding to the perspectives presented in section 2b, WP4 focuses on synthesis. WPs 5-8 support research and innovation activities of the Center (Figure 1 and Table 1).

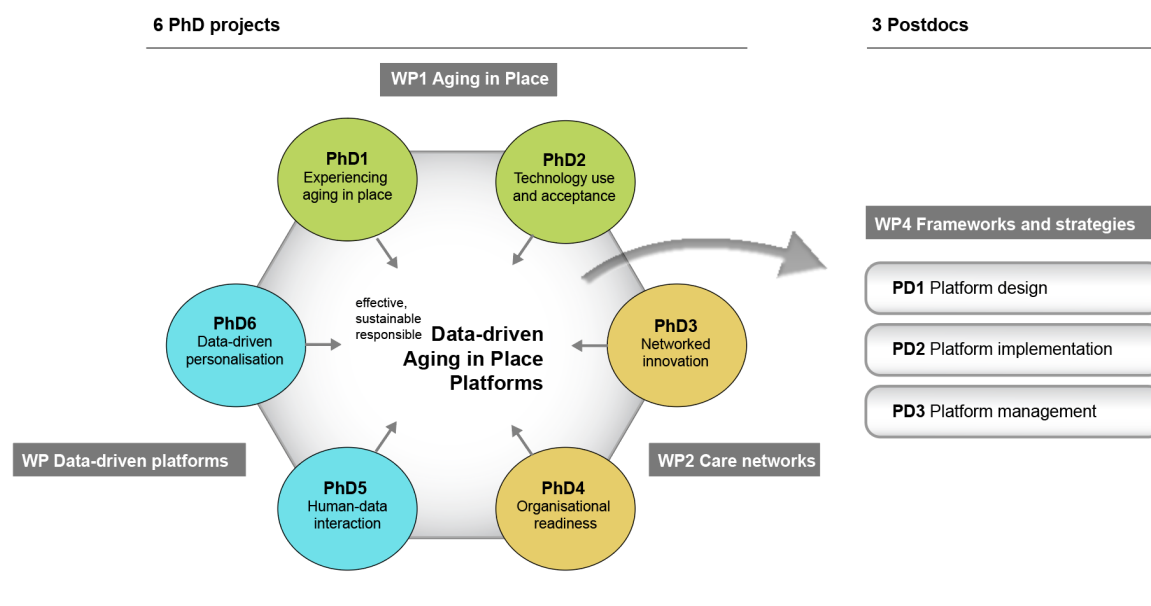


Figure 1: Schematic representation of the work packages, PhD projects and postdoc projects.

WP1: Aging in Place focuses on the experience of AiP. We foreground the perspective of aging adults within their social contexts, while also including quantitative insights on the challenges of aging adults and supporting structures. **PhD1** develops a holistic understanding¹¹ of AiP by mapping out which needs and values are at play during the aging process^{8,27,31}, also incorporating the measurement and model-based usage of data-driven platforms. **PhD2** focuses on the acceptance of data-driven AiP platforms in daily life^{18,19,25}, and how interactions with these platforms as sociotechnical systems change the way care is experienced.

WP2: Care Networks focuses on formal care networks that support AiP and the transformations that data-driven AiP platforms demand from these networks, when they embed technology in their care practices. **PhD3** analyzes stakeholder networks behind platforms (including care professionals, managers, technologists and policy makers) to investigate and anticipate the changing roles and responsibilities of network participants²⁰. **PhD4** researches the readiness of organizations to embed data-driven platforms in their structure and activities^{10,22,28}. It focuses on potential efficiency gains from data-driven AiP platforms and on the implications on workflows, skill requirements, professional norms and quality of care²².

WP3: Data-driven Platforms focuses on the data collection process of data-driven AiP platforms through interconnected products and services and on obtaining data-driven insights enabling personalized and anticipatory support and care. **PhD5** researches data

collection strategies of current data-driven AiP platforms concerning the type and structure of data collected, interactions with data by aging adults and caregivers, the sharing of data within care networks, and the ethical considerations this brings forth⁵. **PhD6** investigates the means in which data-driven insights can be obtained about the adoption, continued usage and effectiveness of the products and services connected to a platform to ensure the aging adults receive the best matching support from the available technologies on a platform over time^{26,30}.

WP4: Frameworks and strategies focuses on conceptualizing data-driven AiP platforms and distilling strategies to design, implement and manage them by synthesizing the knowledge developed in WP1-3. **PD1** focuses on developing an integrated, emancipatory understanding of the design of data-driven AiP platforms. **PD2** focuses on developing routes for implementing data-driven AiP platforms by taking into consideration the transformative effects these platforms have on the way care is experienced, organized, and financed. **PD3** focuses on ensuring the continued use of data-driven platforms by developing smart mechanisms that orchestrate different platform configurations over time.

WP5: Education starts connecting existing courses across the three participating universities. It also aims to develop new interdisciplinary courses: Minors, mixed graduation teams, and developing innovative education formats such as MOOCs (also targeting professional audiences).

Through research projects, design challenges and hackathons, BA and MA students are involved in the Flagship project in all three universities. This may range from Bachelor to Master courses, graduation projects and Minors (existing ones and to-be-developed ones based on the work conducted at the Center). Its management requires the identification of topics and timely embedding within various education schedules and projected student deliverables that feed back into the research and the consortium (section 3c). The Flagship will focus on a close integration of student work aligned with the research. Across collaboration levels, opportunities for education involvement will be monitored.

WP6: Academic talent development fosters a coaching and talent development process to support postdocs in their development towards a tenure track and to recruit new tenure trackers in this highly interdisciplinary field. Our knowledge and innovation center will also invite international guests for research visits to generate international visibility for the junior researchers.

WP7: Knowledge dissemination promotes the visibility and long-term societal impact of our Flagship's knowledge and innovation center by establishing a strong online presence (website, social media), organizing regular partner meetups, creating opportunities for media coverage, including regular television appearances with our partner Omroep MAX.

WP8: Project management includes managing main activities of the center (section 4d), arranging the facilities required and nurturing the consortium as a vibrant community.

Required facilities and data-and infrastructure plan. Having a shared physical workspace to work and meet regularly is important given the multidisciplinary nature of the Flagship. We will look into the availability of a suitable working environment similar to creative co-working spaces such as the ID-Studiolab at TU Delft or Convergence Square

at EMC. Given our focus on innovation, physical and digital prototyping facilities are important resources. We have access to prototyping facilities at TU Delft including the ID-Studiolab, AppliedLabs, DEMO and maker space at the Delft Science Center. These labs and spaces also provide technical support. Local ad-hoc networks and infrastructures to experiment with can be set-up with support of our ICT-departments. However, when deploying prototypes and studies in the field with partners, we will be using their networks and infrastructures. Thus, support by data stewards and our legal departments is essential when working with partners in a consortium.

Planning

The projects are distributed over the five years as follows (Table 1). We start in WP1-3 with PhD projects that take stock of current practices and finetune the project scope (PhD1,3,5). One year later, we start the projects with a deepening character (PhD2,4,6). Similarly the postdoc projects vary in starting dates; PD1 takes an integrated and emancipatory approach to develop a framework for data-driven AiP design requirements, PD2 will develop strategies for their implementation and PD3 will create data-driven mechanisms for effective, sustainable and responsible platforms.

Table 1. List of research-oriented WPs, topics, projects, supervisory teams and complementary expertise (next page).

		2023	2024	2025	2026	2027	PhD supervisory teams	Complementary expertise
WP1	PhD1	Experiencing ageing in place					EUR + TUD School of Health Policy and Management (EUR), Technology, Policy and Management (TUD)	EMC Geriatric Medicine (EMC)
	PhD2	Technology use and acceptance					TUD + EMC Industrial Design Engineering (TUD), Geriatric Medicine (EMC)	EUR School of Health Policy and Management (EUR)
WP2	PhD3	Networked innovation					EMC + EUR Public Health + General Practitioners (EMC), School of Health Policy and Management (EUR)	TUD Technology, Policy and Management (TUD)
	PhD4	Organisational readiness					EMC + TUD General Practitioners + Public Health (EMC), Industrial Design Engineering (TUD),	EUR School of Health Policy and Management (EUR)
WP3	PhD5	Human-data interaction					TUD + EUR Industrial Design Engineering (TUD), Erasmus School of Economics (EUR)	EMC Geriatric Medicine (EMC)
	PhD6	Data-driven personalisation					EUR + EMC School of Economics (EUR), Public Health (EMC)	TUD Technology, Policy and Management (TUD)
WP4	PD1	Platform design					EMC + EUR Geriatric Medicine (EMC), School of Economics (EUR)	
	PD2	Platform implementation					TUD + EMC Industrial Design Engineering (TUD), Public Health (EMC)	
	PD3	Platform management					EUR + TUD School of Economics (EUR), Industrial Design Engineering (TUD)	

3.d Alignment of the program with relevant national and international research agendas

We see a relevant shift towards AiP which is indicated by the Health and Care mission of the Dutch Ministry of Health, Welfare and Sport (VWS). The urgency of this is highlighted by the newly established position of a Minister of Long-term Care in the current Dutch government. Our ambition to research and innovate data-driven platforms to support AiP fits with these societal and governmental developments. In line with this focus, ZonMW has launched a call for research on Access to Care in the Living Environment (KIC) to which our Flagship team is submitting a project proposal.

The new Horizon Europe strategic plan lists health in its first cluster. This cluster aims to “contribute to safer, trusted, more effective and efficient, affordable and cost-effective tools, technologies and digital solutions for improved (personalized) health promotion and disease prevention.” Our Flagship is aligned with this research agenda. US NIH has launched a call for research on health care models for persons with multiple chronic conditions with a specific focus on technology in relation to self-management, timely continuity of care and coordination in care settings with limited resources and clinical personnel. In conclusion, all developed economies are facing an aging population, therefore, our Flagship aligns with research agendas at a global scale.

4. Consortium

4.a Track record of the consortium members

Appendix IV shows the track-record of the applicants highlighting their expertise, two key publications and notable mentions on funding, research impact and education. We represent different expertise's from 3 universities across different research groups, making this Flagship a unique collaborative project. We have a balanced gender distribution (13 female, 12 male). This team has successfully acquired large national and international grants and has at least 5 members with a strong position in coordinating and renewing education.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The rich interdisciplinary configuration of our consortium allows us to robustly understand data-driven AiP platforms from the three interrelated perspectives described in section 2b.

1. Aging in Place: **Prof. Francesco Mattace-Raso** and **Prof. Tischa van der Cammen** have a background in Geriatric Medicine. Mattace-Raso's knowledge on age-related cardiometabolic changes and the consequences of functional and cognitive decline and van der Cammen's expertise on comprehensive geriatric assessment, ensures work based on a deep medical understanding of AiP. **Dr. Rik Wehrens'** expertise in medical anthropology and ethnographic methodology will help us to articulate the experiential richness of the life-worlds of aging adults in relation to digital and data-driven health and AI. To obtain a quantitative understanding of AiP, **Prof. Bas Donkers** and **Prof. Benedict Dellaert** apply their expertise on econometrics and machine learning to study decision-making and behavior of aging adults by using quantitative research methods. **Prof. Esther de Bekker-Grob** contributes in patients' preferences, choice behavior and economic evaluations. To understand AiP in its broader social and economic context, **Prof. Hein Raat** and **Prof. Lex Burdorf** provide expertise on the socio-demographic characteristics of aging adults, loneliness and the cost-effectiveness of occupational health programmes. **Dr. Jorien Veldwijk's** research covers a broad range of health care and public health topics further enriching our research.

2. Care Networks: To identify how Data-driven AiP platforms are likely to transform the functioning and organization of formal care networks, we have expertise in Health Policy and Management. The complexity and dynamics of care networks, roles of stakeholders and institutions have been studied extensively by **Dr. Iris Wallenburg**, who brings in experience with action-research in the field of older person care, as well as in-depth knowledge on organizational change in health care due to technological innovation. To anticipate the ways in which AiP platforms will generate shifts in responsibilities and require new skills and ways of working we benefit from **Dr. Marleen de Mul**'s research on the normalization of technology in care practices, understood through organizational studies and implementation science. **Dr. Armagan Albayrak**'s researches healthcare systems as sociotechnical systems, identifies how to carry out research in a medical setting and how to communicate with multiple stakeholders and users within a healthcare context. **Prof. Patrick Bindels** and **Dr. Rianne Rozendaal**, have conducted innovative research on eHealth with TUDelft, with expertise on the key role played by GP practices in coordinating care around aging persons.

3. Data-driven Platforms: The design and use of data-driven platforms requires another set of disciplines. **Dr. Marco Rozendaal** and **Prof. Pieter Jan Stappers** bring in expertise in Research-through-Design applied to healthcare contexts and how to organize participation in research around participatory design tools and prototypes. With a background in distributed and multi-agents systems, **Prof. Frances Brazier** sheds light on the design of sociotechnical ecological systems that enable and support participation and empowerment in today's networked society. **Dr. Evangelos Niforatos** and **Prof. Alessandro Bozzon** will apply their expertise of human-computer interaction, artificial intelligence to enable intuitive and meaningful interactions with increasingly intelligent interactive systems.

A guiding aim that runs throughout the above-three perspectives is the development of normative guidelines for a responsible transition towards data-driven AiP platforms. To this end, **Prof. Ibo van de Poel**, **Prof. Sabine Roesser**, and **Dr. Janna van Grunsven** will provide expertise in ethics of technology, including value sensitive design, responsible innovation, risk ethics, and emancipatory design, which they have applied to hotly debated technology-driven transitions in health care.

Education: A further key aim of our program is to embed our research in educational activities. **Dr. Marleen de Mul** is director of the Bachelor program at ESHPM (EUR), responsible for health sciences education in technology & innovation. **Prof. Bas Donkers** coordinates the data science and marketing analytics MSc program at ESE (EUR). **Dr. Janna van Grunsven** coordinates TUD's engineering ethics education as well as the minor Responsible Innovation. **Dr. Armagan Albayrak**, is the coordinator of the "Medisign" specialization at the Faculty of Industrial Design Engineering at TUD, which trains master students to design for the healthcare domain.

4.c Involvement of partners

Our Flagship depends on the collaboration with partners to ground the research objectives in real-world cases that reflect the status quo and are open to scrutiny. This will narrow down our research scope and provide a concrete basis for developing effective, sustainable and responsible data-driven platforms for AiP. Our ecology of complementary stakeholders act as a testbed for developing future platform concepts and improvements. We will collaborate with 5 partner-groups across the different work packages (Table 2).

1. Aging adults partners are involved to include needs, preferences and values of aging adults. **Genero** advocates and supports effective elderly care by participating in health initiatives. **NOOM** is committed to improving the socio-economic status and QoL of aging adults with a migrant background. **EuroQol** develops measuring scales for QoL and

participates in the research to learn how experiences of aging adults can be translated into QoL metrics that can be used in data-driven AiP platforms. **Bureau50** is a marketing and communication agency focusing on 50+ audiences with experience on platform innovation projects for aging adults. Learning about what makes these platforms work is what connects them to the research.

2. Care network partners involve healthcare organizations and networks that deliver social and medical support as people age in place and finance it. **Surplus**, **Aafje** and **Laurens** are healthcare organizations that provide a complete service offering for aging adults. Their contribution to the research is by sharing experiences on digital transitions in healthcare. **Primeur** is a network of General Practitioners (GPs) who, as medical professionals, coordinate care for aging adults living at home. They participate to learn about the value of platforms as an opportunity for eHealth. **CZ** is one of the larger health insurance companies in the Netherlands with an interest to learn about the quality and viability of care through platforms with (big) data.

3. Platform partners represent a group of stakeholders who have developed platforms offering AiP-supporting products and services with varying levels of data granularity, stages of development, and scale. **MobileCare**, **ZoDichtbij** and **Zodos** are organizations that deploy data-driven AiP platforms for 200-15000 active users. Data about aging adults is collected by products connected to the platform and people's preferences and experiences are assessed through online questionnaires, diaries or conversations. Data is stored in the cloud. Aging adults can give permission to share data with third-parties within their care networks. This allows care-services to connect to aging adults more efficiently. **Monuta's Vief** and **Omroep Max** are AiP platforms that have a broader focus on people's QoL and produce media content. All platform partners are keen to learn about the reasons why platforms are successful or not, and to innovate their platforms within the consortium.

4. Product partners develop or provide smart products that connect to data-driven AiP platforms. **TinyBots** produces small friendly care robots that interact via speech. **Corsano Health** produces smart bracelets for continuous health monitoring across multiple vital signs. Both have an interest in participating in research to learn about the new product-service combinations made possible when connecting products to platforms. **Vegro** is a traditional wholesaler of assistive products and technologies for aging adults and others. As their field is undergoing a technology-driven transition, Vegro has a keen interest to learn about the different possibilities for making this transition.

5. Broader impact partners represent a group of stakeholders for whom the outcomes of our research will have broader implications on government, finance and society. The **Dutch Ministry of VWS** is involved to learn how AiP platforms help to keep Dutch citizens healthy as long as possible and promote social participation. **ZN** is a union of eleven health insurers in the Netherlands. The research helps to fulfill their mission of arranging healthcare aimed at promoting good quality care that is affordable and accessible. **Omroep MAX** helps translate the research outcomes to aging adults across the Netherlands via their media channels.

Table 2. Distribution of partner-groups across WPs.

Partner group	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8
1. Aging adults	•				•			
2. Care networks		•			•			
3. Platforms	•	•	•	•	•			
4. Products	•		•		•			
5. Broader impact							•	

4.d Governance, organization and collaboration strategy of the consortium

Our **collaboration strategy** involves managing collaboration across five interconnected levels (Figure 4). This requires careful planning and coordination across different time-intervals because of the alignment of research activities with partners and education in different constellations during the course of the project. A project manager will be made responsible for organizing these interactions with stakeholders at various levels within WP8.

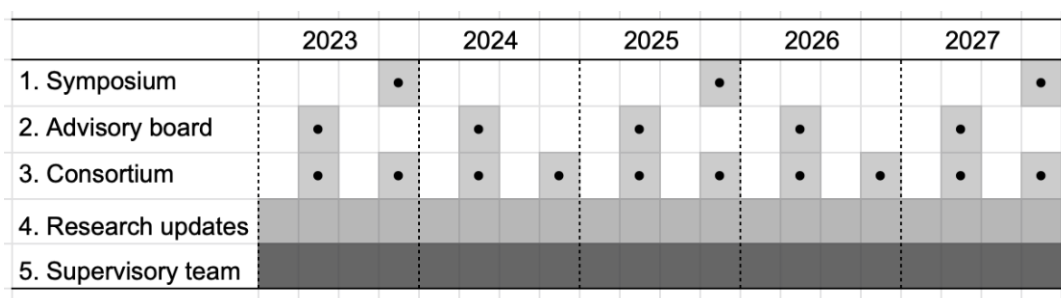


Figure 4. Collaboration strategy across levels over time.

To foster collaboration within the **consortium** we plan meetings every six months where we engage all Flagship partners involving researchers, industry partners and students. During these meetings, knowledge is exchanged and research and innovation opportunities are identified to ensure an ongoing shared momentum on the research project among all partners. An important aspect of these meetings is to finetune the research activities and determine partner involvements for the next stages.

Furthermore, three full-day **symposiums** will be organized. The first symposium will be dedicated to launch the center and present our plans, having inspiring talks from (inter)nationally renowned speakers to show the needs and possibilities to age in place. In the second symposium, we will take stock by connecting with the broader academic and industry community, sharing knowledge and increasing the center's visibility. In the third symposium we will celebrate the center's first lustrum, reflecting on lessons learned in the first episode of the Flagship, and setting the research and innovation agenda for the center's enduring activities.

Our multidisciplinary research ambitions build on a number of **case studies** with multiple researchers, WPs and industry partners involved. Meetings at the level of a single case study will be frequent, as this is where the foundation of our knowledge creation takes place. This requires careful planning and alignment to ensure that relevant stakeholders will stay engaged from conceptualization to execution, analysis and synthesis. This will be the responsibility of Postdocs in close coordination with the project manager (WP8) and

education coordinator when this involves students as well (WP5). The planning for, and the performance of, these case studies will be a recurring topic in the consortium meetings.

To facilitate interdisciplinary collaboration across the case studies and work packages, monthly **research update** sessions will be organized. These two-hour sessions are fast-paced. Each month several researchers present their work to take note, synthesize and reflect collectively on the cross-disciplinary implications of the case studies for the Flagship as a whole. The outcomes will be shared in monthly **flagship meetings** with all researchers involved, i.e. all applicants, newly hired faculty and other interested researchers. These meetings further safeguard the multi-disciplinary perspective required for achieving the Flagship ambitions. These meetings have formal and informal components to optimize knowledge exchange, feedback and network formation. Finally, frequent meetings will occur between the PhDs and their **supervisory teams**.

A **board of directors** will be established for the management of the Flagship project. Their role is to ensure a smooth functioning of the flagship at the intersection of academia and practice. Supported by the project manager, the board will be the first touchpoint for external partners and have a coordinating role in the flagship research projects. To support the board of directors to stay on the right track, an **independent advisory board** will be composed with representatives of the aging adults, care providers, those responsible for financing AiP solutions, the technology and innovation partners, and the academic community. The board of directors will meet with the advisory board annually to report and acquire feedback on the path towards the future.

Annexes

I Budget table

II Letters of intent and letters of commitment

III Letter of support from department head of the main lead

IV Track-record summary table

V Numbered References list

I - Budget table

Flagship:	Aging in Place
Flagship acronym:	AIP
Naam Penvoerder:	TU Delft
Startdatum en looptijd van het project:	december 2022 - december 2027
Totaal project budget:	€ 5.347.900

1.	Looptijden:							2022						2023		2024		2025		2026		2027		Totaal		Total per partner		
	Partner die de kosten maakt																											
	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maanbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoek en ontwikkeling	Delft University of Technology	PhD	2,00	01-12-2022	30-11-2026	P	€ 3.434			€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 494.525	0	€ 494.525
	Delft University of Technology	postdoc	1,00	01-06-2023	31-05-2026	Schaai 10	€ 4.939			€ 44.451	€ 88.902	€ 88.902	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 266.706	0	€ 266.706
	Erasmus University Medical Centre	PhD	2,00	01-12-2022	30-11-2026	P	€ 3.336			€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 120.111	€ 480.444	0	€ 480.444
	Erasmus University Medical Centre	PhD	1,00	01-06-2023	31-05-2026	Schaai 10	€ 4.939			€ 44.451	€ 88.902	€ 88.902	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 266.706	0	€ 266.706
	Erasmus Universiteit	PhD	2,00	01-12-2022	30-11-2026	P	€ 3.434			€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 123.631	€ 494.525	0	€ 494.525
	Erasmus University	postdoc	1,00	01-06-2023	31-05-2026	Schaai 10	€ 4.939			€ 44.451	€ 88.902	€ 88.902	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 44.451	€ 88.902	€ 266.706	0	€ 266.706
	Ondernemingen																											
	Milieu & Natuur		0,00	01-12-2022	30-11-2027		€ 97.500			€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 18.500	€ 97.500	0	€ 97.500
Overige (private) partners	Monika Velt		0,00	01-12-2022	30-11-2027		€ 215.000			€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 43.000	€ 215.000	0	€ 215.000
	Zi dacht		0,00	01-12-2022	30-11-2027		€ 120.000			€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 24.000	€ 120.000	0	€ 120.000
	Bakker		0,00	01-12-2022	30-11-2027		€																					

Partner die de kosten maakt	Materiaal / hulpmiddel	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind
Onderzoeksgenootschappen	Delft University of Technology	platform building	€ 200.000		€ 80.000		€ 80.000		€ 20.000,00		€ 20.000,00		€ 200.000	€ 0
	Delft University of Technology	material costs	€ 200.000		€ 40.000		€ 40.000		€ 40.000		€ 40.000		€ 200.000	€ 0
	Rijksuniversiteit Utrecht	material costs	€ 400.000		€ 40.000		€ 40.000		€ 40.000		€ 40.000		€ 200.000	€ 0
	Eindhoven University Medical Centre	material costs	€ 200.000		€ 40.000		€ 40.000		€ 40.000		€ 40.000		€ 200.000	€ 0
	Rijksuniversiteit Medische Centrum	collaborative spaces	€ 150.000		€ 30.000		€ 30.000		€ 30.000,00		€ 30.000,00		€ 150.000	€ 0
	EUR-ERASMUSCTUD	computational resources	€ 45.000		€ 9.000		€ 9.000		€ 9.000,00		€ 9.000,00		€ 45.000	€ 0
Onderwijsinstellingen	EUR-ERASMUSCTUD	learning fees	€ 100.000		€ 20.000		€ 20.000		€ 20.000,00		€ 20.000,00		€ 100.000	€ 0
	EUR-ERASMUSCTUD	sustainable embedding in research and education	€ 200.000		€ 40.000		€ 40.000		€ 40.000,00		€ 40.000,00		€ 200.000	€ 0
	EUR-ERASMUSCTUD	kosten overbrengen	€ 425.388		€ 85.077,60		€ 85.077,60		€ 85.077,60		€ 85.077,60		€ 425.388	€ 0
	ErasmusZG												€ 0	€ 0,00
	Onderwijsinstelling 2												€ 0,00	€ 0,00
	Onderwijsinstelling 3												€ 0	€ 0,00
Overige (privé) partners	Onderwijsinstelling 4												€ 0	€ 0,00
	Naam 1												€ 0	€ 0,00
	Naam 2												€ 0,00	€ 0,00
	Naam 3												€ 0	€ 0,00
	Naam 4												€ 0	€ 0,00
													€ 0	€ 0,00
Totaal		€ 1.720.388											Totaal kosten materialen en hulpmiddelen	€ 1.720.388

Partner die de kosten maakt	Machine / apparatuur	Prijs per gebruiksaanduiding	Gebruiks-aanduiding	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	Totaal kosten per partner	
Onderzoekorganisaties																		
Delft University of Technology															€ 0,00	€ 0,00	€ 0,00	
Eindhoven University															€ 0,00	€ 0,00	€ 0,00	
Eindhoven University Medical Centre															€ 0,00	€ 0,00	€ 0,00	
EURC/URMCTUD															€ 0,00	€ 0,00	€ 0,00	
Onderaannemers																		
Onderaanneming 1															€ 0,00	€ 0,00	€ 0,00	
Onderaanneming 2															€ 0,00	€ 0,00	€ 0,00	
Onderaanneming 3															€ 0,00	€ 0,00	€ 0,00	
Onderaanneming 4															€ 0,00	€ 0,00	€ 0,00	
Overige (private) partners																		
Naam 1															€ 0,00	€ 0,00	€ 0,00	
Naam 2															€ 0,00	€ 0,00	€ 0,00	
Naam 3															€ 0,00	€ 0,00	€ 0,00	
Naam 4															€ 0,00	€ 0,00	€ 0,00	
Totaal					€ 0												Totaal kosten machines en apparatuur	€ 0

Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind
Orderderes kongressen/afes																
CapX University of Technology															€ 0	€ 0,00
Erasmus University															€ 0	€ 0,00
Erasmus University Medical Centre															€ 0,00	€ 0,00
Organisatie 4															€ 0	€ 0,00
Orderderes vergen																
Orderdering 1															€ 0	€ 0,00
Orderdering 2															€ 0	€ 0,00
Orderdering 3															€ 0,00	€ 0,00
Orderdering 4															€ 0	€ 0,00
Overige (privé) partners																
Naam 1															€ 0	€ 0,00
Naam 2															€ 0,00	€ 0,00
Naam 3															€ 0	€ 0,00
Naam 4															€ 0	€ 0,00
Totaal	€ 0														Totaal kosten derde	€ 0

Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind
Onderzoekskongressies																
Dalhousie / communicate		€ 80.000			€ 80.000				€ 80.000						€ 80.000	€ 0
East University of Technology		€ 80.000			€ 80.000				€ 80.000						€ 80.000	€ 0
Erasmus University		€ 80.000			€ 80.000				€ 80.000						€ 80.000	€ 0
Erasmus University Medical Centre		€ 80.000			€ 80.000				€ 80.000						€ 80.000	€ 0
Eurostars/UMC/UD		€ 125.000			€ 125.000				€ 125.000						€ 125.000	€ 0
Ondernemingen																
Onderneming 1													€ 0		€ 0	€ 0,00
Onderneming 2													€ 0,00		€ 0	€ 0,00
Onderneming 3													€ 0		€ 0	€ 0,00
Onderneming 4													€ 0		€ 0	€ 0,00
Overige (private) partners																
Naam 1															€ 0	€ 0
Naam 2															€ 0	€ 0
Naam 3															€ 0	€ 0
Naam 4															€ 0	€ 0
	Totaal	€ 365.000													Total kosten dusdan	€ 365.000

		2022		2023		2024		2025		2026		2027		Total		per partner	
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind
Onderzoekorganisaties																	
1	Delft University of Technology														€ 0	€ 0	€ 0
2	Eindhoven University of Technology														€ 0	€ 0	€ 0
3	Eindhoven University Medical Centre														€ 0	€ 0	€ 0
4	Organisme 4														€ 0	€ 0	€ 0
Totaal bijdrage onderzoekorganisaties		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Ondernemingen																	
5	Modula Cars	€ 18.900		€ 18.900		€ 18.900		€ 18.900		€ 18.900		€ 18.900			€ 97.500		€ 97.500
6	Modula Vaf	€ 43.000		€ 43.000		€ 43.000		€ 43.000		€ 43.000		€ 43.000			€ 215.000		€ 215.000
9	Zu dotcom	€ 24.000		€ 24.000		€ 24.000		€ 24.000		€ 24.000		€ 24.000			€ 120.000		€ 120.000
10	Zu dotcom	€ 12.000		€ 12.000		€ 12.000		€ 12.000		€ 12.000		€ 12.000			€ 60.000		€ 60.000
8	Yeaan	€ 5.000		€ 5.000		€ 5.000		€ 5.000		€ 5.000		€ 5.000			€ 25.000		€ 25.000
11	Bureau 50	€ 5.880		€ 5.880		€ 5.880		€ 5.880		€ 5.880		€ 5.880			€ 29.400		€ 29.400
10	Coriano	€ 4.000		€ 4.000		€ 4.000		€ 4.000		€ 4.000		€ 4.000			€ 20.000		€ 20.000
11	Chine	€ 2.700		€ 2.700		€ 2.700		€ 2.700		€ 2.700		€ 2.700			€ 13.500		€ 13.500
12	EuroQat	€ 0	€ 200.000	€ 0	€ 200.000	€ 0	€ 200.000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 220.000	€ 0	€ 220.000	€ 0	€ 220.000
13	Truq	€ 3.600		€ 3.600		€ 3.600		€ 3.600		€ 3.600		€ 3.600			€ 18.000		€ 18.000
Totaal bijdrage ondernemingen		€ 0	€ 119.680	€ 200.000	€ 119.680	€ 200.000	€ 119.680	€ 0	€ 119.680	€ 0	€ 119.680	€ 0	€ 0	€ 220.000	€ 598.400	€ 0	€ 818.400
Overige (private) partners																	
14	Brusselsair	€ 5.000		€ 5.000		€ 5.000		€ 5.000		€ 5.000		€ 5.000			€ 25.000		€ 25.000
15	Bruplus	€ 1.000		€ 1.000		€ 1.000		€ 1.000		€ 1.000		€ 1.000			€ 5.000		€ 5.000
16	Bruplus	€ 18.000		€ 18.000		€ 18.000		€ 18.000		€ 18.000		€ 18.000			€ 90.000		€ 90.000
17	Nrc	€ 1.600		€ 1.600		€ 1.600		€ 1.600		€ 1.600		€ 1.600			€ 8.000		€ 8.000
18	ICZ	€ 19.100		€ 19.100		€ 19.100		€ 19.100		€ 19.100		€ 19.100			€ 95.500		€ 95.500
19	Concep Max	€ 20.000		€ 20.000		€ 20.000		€ 20.000		€ 20.000		€ 20.000			€ 100.000		€ 100.000
20	Slamers	€ 10.000		€ 10.000		€ 10.000		€ 10.000		€ 10.000		€ 10.000			€ 50.000		€ 50.000
21	Slamers	€ 1.200		€ 1.200		€ 1.200		€ 1.200		€ 1.200		€ 1.200			€ 6.000		€ 6.000
22	Slamers	€ 1.200		€ 1.200		€ 1.200		€ 1.200		€ 1.200		€ 1.200			€ 6.000		€ 6.000
23	Player van Forest	€ 1.400		€ 1.400		€ 1.400		€ 1.400		€ 1.400		€ 1.400			€ 7.000		€ 7.000
24	VMS	€ 0	€ 100.000	€ 0	€ 100.000	€ 0	€ 100.000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 100.000	€ 0	€ 500.000	€ 0	€ 500.000
Totaal overige bijdrage		€ 0	€ 78.900	€ 100.000	€ 78.900	€ 100.000	€ 78.900	€ 100.000	€ 78.900	€ 100.000	€ 78.900	€ 100.000	€ 0	€ 500.000	€ 594.500	€ 0	€ 894.500
Werkvesting / subsidies / sponsors																	
1	Convergence	€ 200.000		€ 400.000		€ 400.000		€ 400.000		€ 400.000		€ 200.000		€ 200.000	€ 2.000.000	€ 0	€ 2.000.000
2	Werkvesting in care in the home environment	€ 400.000		€ 350.000		€ 350.000		€ 350.000		€ 350.000		€ 350.000		€ 0	€ 1.200.000	€ 0	€ 1.200.000
3	VMS EUR	€ 0		€ 130.000		€ 130.000		€ 130.000		€ 130.000		€ 130.000		€ 200.000	€ 0	€ 260.000	
4	NOE-Intelligence	€ 0		€ 212.500		€ 212.500		€ 212.500		€ 212.500		€ 212.500		€ 425.000	€ 0	€ 850.000	
Totaal		€ 200.000	€ 0	€ 1.092.500	€ 0	€ 1.092.500	€ 0	€ 750.000	€ 0	€ 600.000	€ 0	€ 200.000	€ 0	€ 3.935.000	€ 0	€ 3.935.000	
Totaal budget per jaar		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	Total	Total
		€ 400.000	€ 0	€ 1.581.000	€ 0	€ 1.581.000	€ 0	€ 1.043.500	€ 0	€ 898.500	€ 0	€ 5.547.500	€ 0	€ 5.547.500	€ 0	€ 5.547.500	€ 5.547.500
Totaal budget		€ 368.500	€ 0	€ 1.581.000	€ 0	€ 1.411.000	€ 0	€ 1.043.500	€ 0	€ 898.500	€ 0	€ 5.547.500	€ 0	€ 5.547.500	€ 0	€ 5.547.500	€ 5.547.500

II - Letters of intent and letters of commitment



Marco Rozendaal
Associate Professor of Interaction Design
Delft University of Technology
Landbergstraat 15
2628 CE Delft

Raad van Bestuur

Postbus 8604
3009 AP Rotterdam

Bezoekadres
BrainparkII
Lichtenauerlaan 162-178
3062 ME Rotterdam

T 088 823 1000
I www.aafje.nl

IBAN NL43 ABNA 0248 1733 16
KvK 24388202

DATE March 11 2022

LETTER WITH
REFERENCE 202203009

Subject

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Guy Buck, in my capacity of Chairman of the Board at Stichting Aafje thuiszorg huizen zorghotels, hereby confirm that Stichting Aafje thuiszorg huizen zorghotels intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Stichting Aafje thuiszorg huizen zorghotels intends to provide an in-kind contribution of participating in workshops fostering knowledge exchange, representing a monetary value of € 6000 (80 hours), based on the current information about the flagship. The further detailing of the proposed contribution in the final proposal will be decisive for Stichting Aafje thuiszorg huizen zorghotels in terms of participation and contribution.

Yours sincerely,

Guy Buck

Chairman of the Board

BUREAU VIJFTIG.

Floris Venneman
Dreef 32
2012 HS Haarlem

23th of February, 2022

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

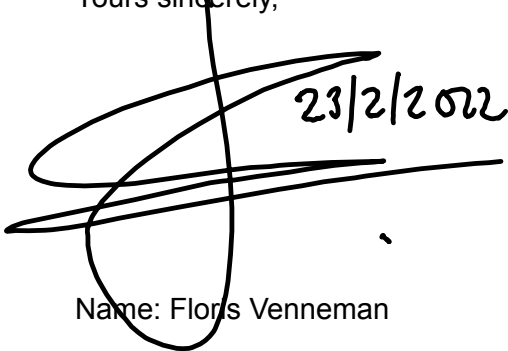
LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Floris Venneman, in my capacity of CEO at Bureauvijftig, hereby confirm that Bureauvijftig intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Bureauvijftig intends to provide an in-kind contribution of quantitative research and commitment of the consultants, representing a monetary value of €29.400,- and further detailed in the project proposal and budget form.

Yours sincerely,

A handwritten signature in black ink, consisting of a large, stylized 'F' with a horizontal line extending to the right. Above the signature, the date '23/2/2022' is written in a cursive, handwritten style.

Name: Floris Venneman

Position: CEO

Date: 23th of February 2022

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

CBRE B.V.
Anthony Fokkerweg 15
1059 CM Amsterdam

+31 (0)20 626 26 91
Netherlands@cbre.com
www.cbre.nl

Kamer van Koophandel
Nr. 33191573
Gereguleerd door RICS

Amsterdam, 14th of March 2022

LETTER OF INTENT for the FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Laura Seckel, in my capacity of Director at CBRE hereby confirm that CBRE B.V. intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

CBRE B.V. intends to provide an in-kind contribution of level 1 of participation: Exchanging knowledge (80 hours) In meetups that are organized two times a year, you can learn about the developments in the Flagship and participate in small workshops fostering knowledge exchange, representing a monetary value of € 13.500,00 and further detailed in the project proposal and budget form.

Yours sincerely,

Laura Seckel
Director Healthcare

Pieter Frederik
Executive Director
Development Services

Marco Rozendaal
Associate Professor of Interaction Design
Delft University of Technology
Landbergstraat 15
2628 CE Delft

8 March 2022

**LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT**

Dear Marco Rozendaal,

I, Peter Stas, in my capacity of CEO at Corsano Health B.V. hereby confirm the intention to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Corsano intends to participate in the project's case studies by providing an in-kind contribution of Level 1 (Exchanging knowledge) representing a monetary value of €20.000 to be further detailed in the project proposal and budget form.

Yours sincerely,



Name: Peter Stas

Position: CEO

Date: 10/3/2022



to:

Technische Universiteit Delft
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft



7 maart 2021

LETTER OF INTENT

for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Menno Jansen, in my capacity of Strategic Program Manager at CZ Zorgverzekeringen hereby confirm that CZ intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

CZ intends to provide an in-kind contribution of participating in action research and the cocreation of innovations, representing a monetary value of € 97.500,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Menno Jansen

Position: Strategic Program Manager

Date: 14 maart 2022

Bas Donkers
Erasmus School of Economics
Erasmus University Rotterdam
Burg. Oudlaan 50, Rotterdam
3062 PA Rotterdam
The Netherlands

EuroQol Research Foundation

Marten Meesweg 107
3068 AV Rotterdam
The Netherlands

T +31 882026890
E userinformationservice@euroqol.org
www.euroqol.org

Date: 10-03-2022

Subject: Letter of intent

Appendix: Your memo "Possible topics of a EuroQol funded project for a postdoc"

Dear Bas,

Thank you for informing us about your planned research in the project "Designing Digital Technologies for Aging in Place: An Interdisciplinary Emancipatory Approach". We appreciate the time and effort that you and your team afforded us in discussing this opportunity and the information that has been provided so far. This message on behalf of the EuroQol Research Foundation ("Foundation") confirms the intent to cofund this research.

The letter of intent is provided for the benefit of the shared objective to further the knowledge about how to assess health and wellbeing outcomes and contribute to the further development of the EQ-HWB, an instrument owned by EuroQol, by doing research along the lines suggested in the appendix. The intent to cofund is based on the following conditions

- A budget for the abovementioned project will only be made available if a budget for the study is secured in the Flagship Program 2022
- The letter of intent shall be non-binding. A formal agreement will be made after you have submitted a full application for a grant via the online submission portal, euroqol-proposals.grantplatform.com, and the Executive committee has decided to award the grant
- The intended award amount in this letter is € 200,000, allowing you to employ a postdoctoral researcher for a period of 2 years, and € 20,000 for data collection for additional validation data.

Finally, the areas of research as described in the appendix are considered of interest for the EuroQol Research Foundation. You may submit a research proposal on your own account, but you also have the option to collect input from the EQ-HWB special interest group to develop a proposal.

Yours sincerely,

Elly Stolk

Scientific Director
EuroQol Research Foundation

To:

Dr. Marco Rozendaal
Main applicant of the Aging in Place project

Location: Rotterdam
Date: March, 9 2022

Subject: Letter of intent to the 'Aging in Place project'

Dear Marco Rozendaal,

With this letter I, Annemiek Beukman, in my capacity as Chairman of GENERO, the network for elderly care in the Southwest Netherlands region, declare that we intend to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

In GENERO, organizations from the southwest of the Netherlands work together to improve care and welfare for (vulnerable) older people. We help to solve bottlenecks that the elderly, their relatives or healthcare professionals identify. We develop improvement initiatives, share knowledge about effective elderly care and support their implementation. We also realize meetings between all kinds of parties in elderly care. Our main driver: we cannot improve elderly care without older people themselves. Therefore, older people play a central role in our network through their participation in the Elderly and Informal Care Forum. They give solicited and non-solicited advice and participate throughout all our activities.

The search for supportive environments that meet both the needs of older people and their informal care-givers is high on the agenda of both our network partners as well as elderly themselves. Therefore, we are keen to contribute to the Aging in place project by searching for technology-based solutions that meet the needs and capabilities of older people.

Provided that the full proposal is granted in the abovementioned call, GENERO will make an in-kind contribution valued at € 50.000 to the project.

Our in-kind contribution will consist of:

- Participating in the 'Market place' throughout the whole Aging in place project (with two persons; one of which will participate in-kind;



- Organisation of, and co-creation in, focus groups to gain knowledge on older people's perspectives on Aging in place;
- Organisation of lunch sessions concerning Aging in Place related themes;
- Assistance in recruiting elderly participants throughout the project, via various channels like newsletters and unions of seniors;
- Contribute to disseminating the knowledge and products to regional organizations in southwest of the Netherlands

Yours sincerely,

Annemiek Beukman
Chairman GENERO
March, 9th, 2022



Letter of Intent PLACEHOLDER LAURENS

Van: Kim Wingerden <kimvanwingerden@laurens.nl>

Datum: vrijdag 11 maart 2022 om 13:23

Aan: Marijke Idema - IO <M.K.Idema@tudelft.nl>

Onderwerp: Intentieverklaring Laurens

Goedemiddag Marijke,

De getekende intentieverklaring vanuit Laurens volgt uiterlijk aankomende dinsdag.
Ondertekening vindt plaats door onze directeur Thuiszorg, Geertje Morée-van Cappellen.

Alvast een goed weekend gewenst.

Met vriendelijke groet,

Kim van Wingerden-Kneepkens
Programmamanager Langer Thuis
Aanwezig: di, woe, don, vrij

Laurens Thuiszorg

M 06-20489817

E kimvanwingerden@laurens.nl

I www.laurens.nl

Omroep MAX
Joop van den Endeplein 9
1217 WJ Hilversum
Postbus 671
1217 WJ Hilversum



Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Jan Slagter, in my capacity of director at public broadcaster Omroep MAX, hereby confirm that Omroep MAX intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Omroep MAX intends to provide an in-kind contribution in several ways, provided that Omroep MAX is the only mediapartner in this project. The contribution can vary in different stages of the project, from bringing in expertise on the usergroup by our staff, defining use cases, addressing the potential users (our members and community – people aged 50 and up – being the core targetgroup of this project), to get input on their needs or for testing of prototypes, to communication with a national reach via our mediaplatforms and perhaps even by addressing the subject Aging in Place in a television programme. This is probably representing a monetary value of € 100.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Jan Slagter
Position: Director Omroep MAX
Date: March 7, 2022



MobileCare BV

Joan Muyskensweg 22

1096CJ Amsterdam

KvK-nummer 68652852

CEO: Patrick Landsman

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Patrick Landsman in my capacity of CEO at MobileCare BV hereby confirm that MobileCare intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

MobileCare intends to provide an in-kind contribution of 1.000 hours support, consulting, access to client data, research etc over a period of 5 years, in average a 200 hours per year in kind representing a monetary value of € 97.500,- (65 euro per hour x 1,5) and further detailed in the project proposal and budget form.

Yours sincerely,

A handwritten signature in black ink, followed by a horizontal line.

Name: Patrick Landsman

Position: CEO

Date: 28-02-2022

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

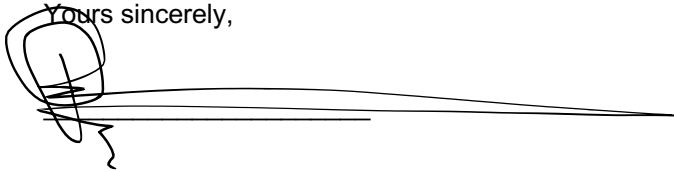
LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Bas Brouwer, in my capacity of Manager Corporate Development at Monuta Holding N.V. hereby confirm that Monuta Ontzorgt B.V. (Vief) intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Monuta Ontzorgt B.V. (Vief) intends to provide an in-kind contribution of meetups, participatory research and co-creating innovations, representing a monetary value of € 215.000 and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Bas Brouwer

Position: Head of Corporate Development, Monuta

Date: 7 March, 2022

Kantoorgebouw Le Mirage - NOOM - Kamer 328
Vliegend Hertlaan 15-97, 3526 KT Utrecht

Volg ons ook op [Twitter](#) en/of [Facebook](#)



Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

Utrecht, february 25 2022

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Bouchaib Saadane, in my capacity of chairman of NOOM at Utrecht hereby confirm that NOOM intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

NOOM intends to provide an in-kind contribution of exchanging knowledge (80 hours) representing a monetary value of € 8.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Bouchaib Saadane

Position: Chairman

Date: February 25, 2022



Pieter van Foreest

Gert-Jan Waterink
Lid Raad van Bestuur
Kalfjeslaan 2
2623 AA Delft
T: 015 515 5315
M: 06-18075906
g.waterink@pietervanforeest.nl

14/03/2022

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Gert Jan Waterink in my capacity of bestuurder at zorginstelling Pieter van Foreest hereby confirm that zorginstelling Pieter van Foreest intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Zorginstelling Pieter van Foreest intends to contribute € 0 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Zorginstelling Pieter van Foreest intends to provide an in-kind contribution of professional expertise, representing a monetary value of € 7000,- and further detailed in the project proposal and budget form.

Letter of Intent for a project partner



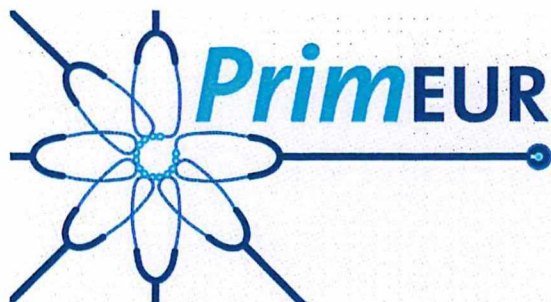
Pieter van Foreest

Yours sincerely,

Name: G.J. Waterink

Position: lid raad van bestuur Pieter van Foreest.

Date: 14/03/2022



March 8th, 2022

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, prof. Patrick Bindels, in my capacity of head of the Department of General Practice at Erasmus Medical Center hereby confirm that Erasmus Medical Center intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Erasmus Medical Center intends to provide an in-kind contribution of 280 hours, representing a monetary value of € 25.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Prof. dr. P.J.E. Bindels

Position: Head of the Department of General Practice

Date: March 8th, 2022

Surplus
Leuvenaarstraat 91,
Postbus 90103
Breda



9-3-2022

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Sandra Suijkerbuijk, in my capacity of programme manager Digital Innovation at Surplus hereby confirm that Surplus intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Surplus intends to provide an in-kind contribution of Exchanging knowledge; 140 hours, representing a monetary value of € 5000,- and further detailed in the project proposal and budget form. Additional activities, including interviews with care staff, will be further discussed within the project and be imbursemented by the project.

Yours sincerely,

A handwritten signature in black ink, appearing to read "Sandra Suijkerbuijk", written over a horizontal line.

Name: Sandra Suijkerbuijk

Position: Programme manager Digital Innovation

Date: 9-3-2022



Van Nelleweg 1
3044BC Rotterdam
info@tinybots.nl

Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

09-03-2022, Rotterdam

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Wang Long Li in my capacity of Managing director at Tinybots hereby confirm that Tinybots BV intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Tinybots intends to provide an in-kind contribution of advising and collaboration hours from personnel, representing a monetary value of € 18.000 and further detailed in the project proposal and budget form.

Yours sincerely,

A handwritten signature in black ink, appearing to read "Wang Long Li", positioned above a horizontal line.

Name: W. L. Li

Position: Managing director

Date: 09-03-2022

Vegro Verpleegartikelen
B.V. Vennestraat 13
2161 LE Lisse



Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

28-2-2022

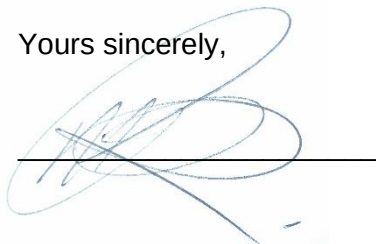
LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Marcel Plomp in my capacity of Innovation Advisor at Vegro Verpleegartikelen B.V. hereby confirm that Vegro Verpleegartikelen B.V. intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Vegro Verpleegartikelen B.V intends to provide an in-kind contribution of Exchanging knowledge and participating in research, representing a monetary value of € 25.000 based upon an average of 0,05 Fte and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Marcel Plomp

Position: Innovation Advisor

Date: 28-2-2022

Technische Universiteit Delft/Industrial Design
Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

14-03-2022

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, Henk Herman Nap, in my capacity of Senior Researcher / Expert eHealth at Vilans, hereby confirm that Vilans intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Vilans intends to provide an in-kind contribution of 8 hours per week (0,2 FTE) for 5 years, representing a monetary value of € 90.000,00 and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Henk Herman Nap

Position: Senior Researcher / Expert eHealth

Date: 14-03-2022

Ministerie van VWS - belangstelling voor Aging in Place en commitment voor kennisuitwisseling

Van: "Kohler-Cornelissen, A. (Anja)" <a.kohler@minvws.nl>

Datum: maandag 14 maart 2022 om 16:43

Aan: Marijke Idema - IO <M.K.Idema@tudelft.nl>

CC: "Pomp, A.G. (Anno)" <ag.pomp@minvws.nl>, "Bleeker, K.A.M. (Karin)" <ka.bleeker@minvws.nl>

Onderwerp: Flagship in place

Dag Marijke,

Graag willen we jullie vriendelijk bedanken voor de toelichting op jullie mooie project. Het gaat hier om betrekkelijk diepgravend onderzoek dat behoorlijke kansen voor de toekomst in zicht heeft.

Naar aanleiding van gesprekken die jullie hebben gehad met mijn collega's Karin Bleeker, Peter Alders en Anno Pomp willen we vanuit het ministerie van VWS laten weten dat we belangstelling uitspreken voor jullie onderzoek. We blijven graag op de hoogte van de vorderingen van het onderzoek. Dat betekent dat we graag periodiek op de hoogte worden gehouden. Ook gaven jullie aan dat het mogelijk is dat we af en toe mee kunnen denken. Daarvan willen we graag gebruik maken.

We wensen jullie alvast heel veel succes bij het onderzoek.

Met vriendelijke groet,

Anja Kohler- Cornelissen

Beleidsmedewerker - Directie langdurige zorg

[Ministerie van Volksgezondheid, Welzijn en Sport](#)

Telefoon: 06-25179795

e-mail: a.kohler@minvws.nl

aanwezig maandag tot en met donderdag

Zo-Dichtbij Medisch Sociale Omgeving bv

e-mail: info@zo-dichtbij.nl
telefoon: 06-81374610
KvK Delft: 82569142
BTW nr.: NL862522201B01
Rek. nr. : NL98 RABO 0367 8190 90
adres. : Dijkshoornseweg 93 2635 EM Den Hoorn



LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

Dear Marco Rozendaal,

I, dr. Wally Keijzer-Broers MBA in my capacity of social entrepreneur at Zo-Dichtbij Medisch Sociale Omgeving MSO hereby confirm that Zo-Dichtbij MSO intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

Zo-Dichtbij MSO intends to provide an in-kind contribution of 800 hours related to the social-technical platform Zo-Dichtbij, as example of a use-case to support people age-in-place, representing a monetary value of € 120.000,-- and further detailed in the project proposal and budget form.

Yours sincerely,



Name: Dr. W.J.W. Keijzer-Broers MBA

Position: Social Entrepreneur

Date: 10-03-2022

De Jonge Consultancy
Wilfred de Jonge
Jensemaheerd 43
9736 CD Groningen



Technische Universiteit Delft/
Industrial Design Engineering
Marco Rozendaal
Landbergstraat 15
2628 CE Delft

LETTER OF INTENT
for the
FLAGSHIP AGING IN PLACE PROJECT

March 11th, 2022

Dear Marco Rozendaal,

I, Wilfred de Jonge, in my capacity of owner at De Jonge Consultancy hereby confirm that De Jonge Consultancy intends to contribute to the Flagship Aging in Place project, as applied for by the main applicant, Marco Rozendaal, associate professor at Human Centered Design at Delft University of Technology.

De Jonge Consultancy intends to provide an in-kind contribution of level 3 participation (Innovating together), representing a monetary value of € 60.000 and further detailed in the project proposal and budget form.

Yours sincerely,

Wilfred de Jonge

Owner

March 11th, 2022

III Letter of support from department head of the main lead

Faculteit Industrieel Ontwerpen

Landbergstraat 15
2628 CE Delft

Delft, March 15th, 2022

Dear members of the Convergence Flagships-committee,

In my capacity as head of the dept. of Human Centred Design at the Faculty of Industrial Design Engineering of Delft University of Technology, I support the submission of the proposal for the Flagship programme 'Aging In Place'.

This collaboration between TU Delft, the University of Rotterdam and Erasmus Medical Centre brings, together with both public and private partners brings great opportunities in both research and innovation on the topic of ageing happy and healthily at home.

With kind regards,



Prof. dr. Ir. P.M.A. Desmet
Head of the department of Human Centred Design

IV - Track-record summary table

Applicant	UNI	Expertise in 3 keywords	2 key references		Notable mentions
Dr. Armagan Albayrak	TUD	design for healthcare, human-centered design, quality of care	Melles, M. et al. (2021). Innovating health care: key characteristics of human-centered design. <i>Int. J. for Qual. in Health Care</i> , 33 (Supplement_1), 37-44.	van der Cammen, T. J. et al. (2019). Where ergonomics meets geriatrics: the connection between comprehensive geriatric assessment.... <i>Eur. Geriatric Med.</i> 10(3), 333-335.	Coordinator of the master specialisation Medisign at Industrial Design Engineering. Education coordinator Benefit fellowship program EIT Health.
Dr. Mariëlle Beenackers	EMC	public health, health inequalities, urban health	Noordzij, J. M., Beenackers, M. A., Diez Roux, A. V., & van Lenthe, F. J. (2019). Age-friendly cities: challenges for future research. <i>Bulletin of the World Health Organization</i> , 97(6), 436–437. https://doi.org/10.2471/BLT.18.224865	Timmermans, E., et al. (2021). Social and physical neighbourhood characteristics and loneliness among older adults: results from the MINDMAP project. <i>J Epidemiol Community Health</i> , 75(5), 464-469.	Coordinator of CEPHIR, the Centre of Effective Public Health In the larger Rotterdam area. Veni laureate. PI in Marie Skłodowska-Curie Innovative Training Network SURREAL and ZonMW projects.
Prof. Esther de Bekker-Grob	EUR	patient preferences, choice behaviour, economic evaluations	de Bekker-Grob, E. W., Ryan, M., & Gerard, K. (2012). Discrete choice experiments in health economics: a review of the literature. <i>Health economics</i> , 21(2), 145-172.	Kamphuis, C. B., et al. (2015). Factors affecting food choices of older adults from high and low socioeconomic groups: a discrete choice experiment. <i>The American journal of clinical nutrition</i> , 101(4), 768-774.	>5MEuro research funding, (NWO Veni & Vidi laureate). editor of the book 'Methods of Health Preference Research', and awarded with Best PhD Supervisor (Erasmus Graduate School).
Prof. Patrick Bindels	EMC	General Practice, family medicine, epidemiology	de Leeuw, M. J., et al. (2022). Healthy Ageing and Intellectual Disability study: summary of findings and the protocol for the 10-year follow-up study. <i>BMJ open</i> , 12(2), e053499.	van Berkel, A. C., et al. (2021). 10-Year natural course of early hip osteoarthritis in middle-aged persons with hip pain: a CHECK study. <i>Annals of the rheumatic diseases</i> , 80 (4), 487-493.	Head of dept General Practice Erasmus MC. He was the chair of the Dutch Health Council committee on the care for frail elderly. He has extensive research experience and conducted several large trials and cohort studies in General Practice.
Prof. Alessandro Bozzon	TUD	AI-enabled systems, user modelling, machine learning	Bozzon, A., et al., (2013, March). Choosing the right crowd: expert finding in social networks. In <i>Proceedings of the 16th international conference on extending database technology</i> (pp. 637-648).	Yang, J. et al.. (2014, September). Asking the right question in collaborative q&a systems. In <i>Proceedings of the 25th ACM conference on Hypertext and social media</i> (pp. 179-189).	Head of Department Sustainable Design Engineering. >5M EUR funding (industry, Dutch and EU), Urban Data Science program manager at AMS Institute.
Prof. Frances Brazier	TUD	socio-technical ecological systems, multi-agents systems, distributed systems	Brazier, F. M., et al. (2009). Agents and service-oriented computing for autonomic computing: A research agenda. <i>IEEE Internet Computing</i> , 13(3), 82-87.	Wijngaards, N. J., et al. (2002). Supporting internet-scale multi-agent systems. <i>Data & Knowledge Engineering</i> , 41(2-3), 229-245.	Multiple Grants obtained including EU FP7 STREP project CIVIS: Cities as drivers of social change. The DST-NWO joint research project on "Smart Grids", ADREM: Adaptive clustering for Decentralized Resilient Energy Management.
Prof. Lex Burdorf	EMC	Public health, epidemiologist, occupational health	Rongen, A., et al. (2013). Workplace health promotion: a meta-analysis of effectiveness. <i>American journal of preventive medicine</i> , 44 (4), 406-415.	Burdorf, A. (2015). Understanding the role of work in socioeconomic health inequalities. <i>Scandinavian Journal of Work, Environment & Health</i> , 41(4), 325-327.	Head of dept Public Health, NIHES and NSPOH teaching, Editor-in-chief <i>Scandinavian Journal of Work</i> .
Prof. Tischa J.M. van der Cammen	TUD	design for aging, falls prevention, integrated care	Jellema, A. H. et al.. (2019). Shoe design for older adults: evidence from a systematic review on the elements of optimal footwear. <i>Maturitas</i> , 127, 64-81	Hartholt et al. (2018). Mortality from falls in Dutch adults 80 years and older, 2000-2016. <i>Jama</i> , 319(13), 1380-1382.	Honorary research fellow in Geriatrics, Erasmus MC. Emeritus Professor on Healthy Aging at Industrial Design Engineering at TU Delft.

Prof Benedict G.C. Dellaert	EUR	(online) consumer decision-making, choice architecture, consumer-firm interactions	Lim, S. et al (2021) - Digital customization of consumer investments in multiple funds: virtual integration improves risk–return decisions. J. of the Acad. of Market. Sc. 49 (4), 723-742	Dellaert, B.C.G. et al. (2020). Consumer Decisions with Artificially Intelligent Voice Assistants. Marketing Letters: a journal of research in marketing.	Multiple grants on supporting (financial) decision making funded by Studies on Pensions, Aging and Retirement (Netspar). Developed new Masters courses on Data Science in an organizational and societal context and on consumers use of digital and physical media channels.
Prof. Bas Donkers	EUR	data-driven understanding of human behavior, econometrics, machine learning	Donkers, B. et al. (2020) - Preference Dynamics in Sequential Consumer Choice with Defaults Opent extern - J. of Market. Research, 57 (6), 1096-1112	Broekhuizen, T.L.J. et al. (2021) Digital platform openness: Drivers, dimensions and outcomes, Journal of Business Research, 122, 902-914	2M EUR (Euroqol, Netspar, Research Excellence). Coordinator MSc program Data Science and Marketing Analytics.
Dr. Jeannette Goudzwaard	EMC	Frailty and functional decline, cardiothoracic interventions in elderly, delirium	Goudzwaard et al (2020). Impact of Frailty on health-related Quality of Life one year after Transcatheter Aortic Valve Implantation. Age and Ageing, 2020, DOI: 10.1093	Goudzwaard et al (2020) Incidence, determinants and consequences of delirium in older patients after transcatheter aortic valve implantation. Age Ageing. April 2020 27;49(3): 389-394	Alzheimer Centrum Erasmus MC, Implementation of Innovative education methods for nurses in the education of older hospitalized patients, supervising. Active member as chair of 2 commissions of Dutch Geriatric Society.
Dr. Amy van Grieken	EMC	Public health, integrated care, community intervention and evaluation studies	Franse, C. B., et al. (2017). Evaluation design of urban health Centres Europe (UHCE): preventive integrated health and social care for community-dwelling older persons in five European cities. BMC geriatrics, 17(1), 1-8.	Van Grieken, A., et al (2017). Personalized web-based advice in combination with well-child visits to prevent overweight in young children: cluster randomized controlled trial. J Med Internet Res, 19(7).	>1M EUR including EU and ZonMw funding (EU: H2020 WELLBASED, ValueCare, K-HiA, ZonMw: PI SQUEALED)
Dr. Janna van Grunsven	TUD	ethics, engineering ethics education, embodied embedded cognition	Van Grunsven, J., & Roeser, S. (2022). AAC technology, autism, and the empathic turn. Soc. Epistem., 36(1), 95-110.	van Grunsven, J. (2021). Perceptual breakdown during a global pandemic: introducing phenomenological insights... Ethics and Inform. Tech., 23(1), 91-98.	PI on 4TU.CEE engineering ethics education research project. Fulbright Scholar. coordinates TUD's engineering ethics education as well as the minor Responsible Innovation.
Prof. Francesco Mattace-Raso	EMC	cardiometabolic change, functional decline, cognitive decline	Akgun-Citak E et al. (2020). Challenges and needs of informal caregivers in elderly care: Qualitative research in four European countries, the TRACE project. Arch Gerontol Geriatr. 2020;87:103971.	Fierloos, I. N. et al. (2021). Socio-demographic characteristics associated with emotional and social loneliness among older adults. BMC geriatrics, 21(1), 1-10.	Head of the Division of Geriatric Medicine. Member European Boards (COST CONNECTION, EuGMS). PI in 4 EU projects.
Dr. Marleen de Mul	EUR	ambient assistive living, social robotics, health information technology	Vandekerckhove, P. et al. (2020). Generative participatory design methodology to develop electronic health interventions: systematic literature review. Journal of medical Internet research, 22(4)	Fiorini, L. et al. (2021). Assistive robots to improve the independent living of older persons: results from a needs study. Disability and Rehabilitation: Assistive Technology, 16 (1), 92-102.	European-Japanese consortia (ICT-PSP, Horizon 2020). Responsible for health sciences education in technology & innovation. Director of the Bachelor programme at ESHPM (EUR)
Dr. Evangelos Niforatos	TUD	human AI systems, human technology augmentation, human-computer interaction	Niforatos, E. et al. (2020, April). Would you do it?: enacting moral dilemmas in virtual reality for understanding ethical decision-making. In Proc. of the 2020 CHI conference on human factors in computing systems (pp. 1-12).	Sharma, K. et al. (2020). Assessing cognitive performance using physiological and facial features: Generalizing across contexts. Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 4(3), 1-41.	>800k EUR in funding (FET Open H2020, SwissNSF and NWO projects). Director Delft AI lab & COALA H2020 project (data-driven argiculture).
Prof. Ibo van de Poel	TUD	responsible innovation, design for values, ethics of newly emerging technologies	Van de Poel, I. (2021). Design for value change. Ethics and Information Technology, 23(1), 27-31.	van de Poel, I. (2020). Embedding values in artificial intelligence (AI) systems. Minds and Machines, 30(3), 385-409.	VICI, ERC Advanced Grant, & NWO-Gravitation Grant.
Prof. Hein Raat	EMC	Public health, PROMs, m-health, vulnerable populations	Feng, Z., et al. (2017). Risk factors and protective factors associated with incident or increase of frailty among community-dwelling older adults: A systematic review of longitudinal studies. PloS one, 12(6)	Franse, C. B., et al.(2017). Socioeconomic inequalities in frailty and frailty components among community-dwelling older citizens. PloS one, 12(11),	Physician Society and Health. Adjunct professor at the School of Public Health of Peking University, Beijing. Visiting professor at the West China School of Public Health of Sichuan University, Chengdu. 6M EUR for PI EU H2020 ValueCare project.

Prof. Sabine Roeser	TUD	risk ethics, moral intuitionism, research integrity	Steinert, S., & Roeser, S. (2020). Emotions, values and technology: Illuminating the blind spots. <i>J. of Resp. Innov.</i> , 7(3), 298-319.	Roeser S. (2018), <i>Risk, Technology, and Moral Emotions</i> , Routledge	>26M Euro including Veni, Vidi, NWO-MVI, & NWO-Gravitation grants from the Dutch Research Council (NWO).
Dr. Marco Rozendaal	TUD	interaction design, research through design, human-computer interaction	Rozendaal, M. C., et al.. (2019). Objects with intent: Designing everyday things as collaborative partners. <i>ACM Transactions on Computer-Human Interaction (TOCHI)</i> , 26(4), 1-33.	Rozendaal, Marco C., Betti Marenko, and William Odom, eds. <i>Designing Smart Objects in Everyday Life: Intelligences, Agencies, Ecologies</i> . Bloomsbury Publishing, 2021.	>1M EUR including 2 NWO Take-off grants, Meedoen=Groeien! Gelijke kansen voor kinderen met kanker (HandicapNL). NWO – KIEM Creatieve industrie on wearable technology for increased body awareness.
Dr. Rianne Rozendaal	EMC	General Practice, epidemiology, quality of care	Arslan, I. G., et al. (2021). Quality indicators for knee and hip osteoarthritis care: a systematic review. <i>RMD open</i> , 7(2), e001590.	Arslan, I. G., et al., (2021). Evaluation of intermediate care for knee and hip osteoarthritis: a mixed-methods study. <i>BMC Family Practice</i> , 22(1), 1-10.	Coordinator of the academic GP network Primeur, with over 90 GPs in the Rijnmond area, that collaborate closely with the department of General Practice in the field of education, vocational training of GPs and research.
Prof. Pieter Jan Stappers	TUD	Design techniques, early concept prototyping, participatory design	Voûte, E. et al. (2020). Innovating a large design education program at a university of technology. <i>She Ji: The Journal of Design, Economics, and Innovation</i> , 6(1), 50-66.	Stappers, P. J., & Giaccardi, E. (2017). Research through design. In <i>The encyclopedia of human-computer interaction</i> (pp. 1-94). The Interaction Design Foundation.	Head of Department: Design Conceptualization and Communication (DCC). Director Faculty Graduate School Industrial Design Engineering (2012-2020).
Dr. Jorien Veldwijk	EUR	clinical epidemiology, public health, discrete choice experiment	Barte, J., et al. (2014). Differences in weight loss across different BMI classes: a meta-analysis... In. <i>J. of Beh. Med.</i> , 21(5), 784-793.	Russo, S. et al. (2019). Understanding patients' preferences: a systematic review of psychological instruments used in patients' preference and decision studies. <i>Value in Health</i> , 22(4), 491-501.	2M EUR funded for 2 Innovative Medicines Initiative (IMIs): EU public-private partnership funding health research and innovation.
Dr. Iris Wallenburg	EUR	action-research, sociology of care and technology, healthcare workforce	Ivanova, D. et al.. (2016). Care in place: A case study of assembling a carescape. <i>Sociology of Health & Illness</i> , 38(8), 1336-1349.	Wallenburg, I., & Bal, R. (2019). The gaming healthcare practitioner: How practices of datafication and gamification reconfigure care. <i>Health Inform. J.</i> , 25(3), 549-557.	>10M EUR funded for multiple projects including: Societal Impact of AI (AiPact), Regional development of elderly care (RegioZ), Nursing practise project (RN2Blend).
Dr. Rik Wehrens	EUR	science and technology, medical anthropology, empirical ethics	Stevens, M., et al. (2022). Why Personal Dreams Matter: How professionals affectively engage with the promises surrounding data-driven healthcare in Europe. <i>Big Data & Society</i> , 9(1)	Wehrens, R. et al.. (2021). Ethics as discursive work: The role of ethical framing in the promissory future of data-driven healthcare technologies. <i>Science, Technology, & Human Values</i>	Artificial Intelligence in Medicine. Healthcare Policy and Management innovation grant.

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